$S°(298.15\ K)$ = [212.652] $J·K^{-1}·mol^{-1}$

$\Delta_f H°(0\ K)$ = [112.3 ± 96.2] $kJ·mol^{-1}$
$\Delta_f H°(298.15\ K)$ = [108.5 ± 96.2] $kJ·mol^{-1}$

Vibrational Frequencies and Degeneracies ν, cm^{-1}
[2327](1)
[1020](1)
[2322](1)

Ground State Quantum Weight: [2] σ = [2]
Point Group = [C_{2v}]
Bond Distance: P–H = [1.419] Å
Bond Angle: H–P–H = [93.7]°
Product of the Moments of Inertia: $I_A I_B I_C$ = [0.0695 × 10^{-117}] $g^3·cm^6$

Enthalpy of Formation

$\Delta_f H°$(298.15 K) was estimated by DiStefano *et al.*[1]. The $\Delta_f H°$(298.15 K) was estimated by taking 63% of the dissociation energy of PH_3 and using it as the dissociation energy of PH_2. Willes and Winkler[2] have estimated $\Delta_f H°(PH_2, g, 298.15\ K)$ to be 30 $kcal·mol^{-1}$.

Heat Capacity and Entropy

The vibratonal spectra of PH_2 from the flash photolysis of PH_3 has been reported by Ramsey[3] and in.[4] Similarities between the spectra of NH_2 and PH_2 were found. For NH_2 and NH_3 bond angles and bond distances were reported to be about the same. Byanalogy then the bond angles and bond distances of PH_2 and PH_3 should be nearly the same. The moments of inertia for PH_2 were therefore calculated from the molecular constants for PH_3. The principal moments of inertia are: I_A = 0.2960 × 10^{-39}, I_B = 0.3588 × 10^{-39}, and I_C = 0.6548 × 10^{-39} $g·cm^2$.

DiStefano *et al.*[1] have assumed the ground state degeneracy to be 2 because of the unpaired electron. The vibrational frequencies reported by DiStefano *et al.* have been used. They used a simple force field and the force constants of PH_3 to obtain the vibrational frequencies.

References

[1]V. N. DiStefano, R. L. Potter and S. N. Fox, "The Thermodynamic Functions of Some Combustion Products Containing Phosphorus–I.", American Cyanamid Company.
[2]D. Willes and C. Winkler, J. Phys. Chem. 61, 902 (1957).
[3]D A. Ramsey, Nature 178, 375 (1956).
[4]N. Y. Acad. Sci. 67, 485 (1957).

Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = $p°$ = 0.1 MPa

T/K	$C_p°$	$S°$	$-[G°-H°(T_r)]/T$	$H°-H°(T_r)$	$\Delta_f H°$	$\Delta_f G°$	log K_f
	$J·K^{-1}·mol^{-1}$			$kJ·mol^{-1}$			
0	.000	.000	INFINITE	−10.006	112.321	112.321	INFINITE
100	33.258	175.957	242.760	−6.680	111.760	106.006	−55.372
200	33.549	199.055	215.787	−3.347	110.136	100.633	−26.283
250	34.075	206.593	213.221	−1.657	109.338	98.348	−20.549
298.15	34.774	212.652	212.652	.000	108.499	96.306	−16.872
300	34.804	212.868	212.653	.064	108.466	96.231	−16.755
350	35.641	218.295	213.080	1.825	106.842	94.334	−14.079
400	36.535	223.112	214.038	3.629	105.873	92.614	−12.094
450	37.468	227.468	215.292	5.479	104.947	91.012	−10.564
500	38.430	231.466	216.712	7.377	104.065	89.512	−9.351
600	40.400	238.646	219.783	11.318	102.445	86.756	−7.553
700	42.343	245.020	222.941	15.456	101.012	84.257	−6.287
800	44.169	250 796	226.067	19.783	99.754	81.952	−5.351
900	45.820	256.095	229.113	24.284	98.648	79.794	−4.631
1000	47.278	261.000	232.060	28.940	97.668	77.753	−4.061
1100	48.548	265.567	234.901	33.733	96.789	75.805	−3.600
1200	49.645	269.839	237.636	38.644	32.417	75.011	−3.265
1300	50.590	273.851	240.269	43.657	32.452	78.560	−3.157
1400	51.403	277.631	242.804	48.758	32.525	82.104	−3.063
1500	52.104	281.202	245.246	53.934	32.627	85.642	−2.982
1600	52.711	284.584	247.600	59.175	32.747	89.172	−2.911
1700	53.238	287.796	249.871	64.473	32.878	92.695	−2.848
1800	53 696	290.852	252.063	69.821	33.015	96.210	−2.792
1900	54.097	293.767	254.182	75.211	33.153	99.717	−2.741
2000	54.450	296.551	256.231	80.638	33.290	103.217	−2.696
2100	54.760	299.215	258.215	86.099	33.422	106.710	−2.654
2200	55 035	301.769	260.137	91.589	33.548	110.197	−2.616
2300	55.280	304.221	262.001	97.105	33.665	113.678	−2.582
2400	55 498	306.578	263.810	102.644	33 774	117.154	−2.550
2500	55.693	308.848	265.566	108.204	33.873	120.627	−2.520
2600	55.869	311 036	267.273	113.782	33.961	124.095	−2.493
2700	56.027	313.147	268.933	119.377	34.037	127.560	−2.468
2800	56 170	315.187	270.549	124.987	34.102	131.023	−2.444
2900	56.299	317.161	272.122	130.611	34.154	134.483	−2.422
3000	56.417	319.071	273.656	136.247	34 194	137.942	−2.402
3100	56.524	320.923	275.151	141.894	34.221	141.400	−2.383
3200	56.622	322.719	276.609	147.551	34.235	144.857	−2.365
3300	56.712	324.463	278.033	153.218	34.236	148.314	−2.348
3400	56.795	326.157	279.424	158.893	34.222	151.771	−2.332
3500	56.871	327.805	280.783	164.577	34.195	155.229	−2.317
3600	56.941	329.408	282.111	170.267	34.154	158.687	−2.302
3700	57.005	330.969	283.411	175.965	34.098	162.147	−2.289
3800	57.065	332.490	284.682	181.668	34.028	165.609	−2.276
3900	57.121	333.973	285.927	187.377	33.944	169.073	−2.264
4000	57.172	335 419	287.146	193.092	33.844	172.539	−2.253
4100	57.220	336.832	288.341	198.812	33.729	176.008	−2.242
4200	57.265	338.211	289.512	204.536	33.600	179.479	−2.232
4300	57.307	339.559	290.660	210.265	33.455	182.954	−2.222
4400	57.346	340.877	291.787	215.997	33.295	186.433	−2.213
4500	57.383	342.166	292.892	221.734	33.121	189.915	−2.204
4600	57.417	343.428	293.977	227.474	32.931	193.401	−2.196
4700	57.449	344.663	295.042	233.217	32.726	196.892	−2.188
4800	57 480	345.873	296.089	238.964	32.507	200.388	−2.181
4900	57.508	347.058	297.117	244.713	32.274	203.887	−2.173
5000	57.535	348.220	298.127	250.465	32.027	207.392	−2.167
5100	57.560	349.360	299.121	256.220	31.766	210.902	−2.160
5200	57.584	350.478	300.098	261.977	31.491	214.417	−2.154
5300	57.607	351.575	301.059	267.737	31.205	217.938	−2.148
5400	57.628	352.652	302.004	273.499	30.906	221.464	−2.142
5500	57.649	353 710	302.935	279.263	30.596	224.995	−2.137
5600	57.668	354.749	303.851	285.028	30.275	228.533	−2.132
5700	57.686	355.769	304.753	290.796	29.944	232.076	−2.127
5800	57.703	356 773	305.641	296.566	29.604	235.626	−2.122
5900	57 720	357 759	306.516	302.337	29.255	239 180	−2.118
6000	57 736	358.730	307.378	308.110	28.899	242.741	−2.113

PREVIOUS March 1963 (1 atm) CURRENT. March 1963 (1 bar)